ABSTRACT

Indian agriculture still depends largely on manual observation of field and weather conditions, which is slow, irregular and often detects crop stress only after visible damage has occurred. Existing precision-agriculture solutions rely on expensive IoT sensor kits that require installation and maintenance, making them unaffordable for small and marginal farmers. This project, AgriSense, presents a software-only system for real-time monitoring of environmental and crop health parameters that requires no hardware investment.

The system acquires live agro-meteorological data - air temperature, relative humidity, rainfall, wind speed, soil moisture and soil temperature - from a satellite-validated weather data service (Open-Meteo API) for the selected farm location. A rule-based health model compares these parameters against crop-specific optimal ranges (wheat, rice, maize, cotton, sugarcane and mustard) and computes a Crop Health Score (0-100), an estimated vegetation index (NDVI), and real-time alerts and recommendations such as irrigation requirement, heat or cold stress, fungal-disease risk, high-wind spray warnings and rainfall forecasts. The web-based dashboard, developed using HTML, CSS, JavaScript and Chart.js, refreshes automatically every 60 seconds and visualises live readings along with past 24-hour trends and 24-hour forecasts.

Testing across multiple agricultural locations of India demonstrates that the system delivers continuous, crop-specific decision support at zero hardware cost, making the benefits of precision agriculture accessible to every farmer. The architecture follows the standard monitoring pipeline (data acquisition, processing, dashboard), so physical field sensors can be integrated in the future without changing the software system.

Keywords: Precision Agriculture; Real-Time Monitoring; Crop Health Score; Soil Moisture; NDVI; Weather API; Rule-Based Alert System; Web Dashboard